

# Homework 5 Math 210A

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October 16, 2025

**Problem 1.** Let  $\varphi : G_1 \rightarrow G_2$  be a group homomorphism.

(a) Show that if  $N_2 \trianglelefteq G_2$ , then  $\varphi^{-1}(N_2) \trianglelefteq G_1$ . Deduce that  $\ker(\varphi) \trianglelefteq G_1$ .

**Proof.** Suppose  $N_2 \trianglelefteq G_2$ . Our goal is to show that  $\varphi^{-1}(N_2) \trianglelefteq G_1$ ; that is, for all  $g \in G_1$ ,

$$g\varphi^{-1}(N_2)g^{-1} \subseteq \varphi^{-1}(N_2). \quad (*)$$

To this end, let  $g \in G_1$  be arbitrary. To show (\*), let  $x \in g\varphi^{-1}(N_2)g^{-1}$  be arbitrary. Then

$$x = g\tilde{x}g^{-1} \text{ for some } \tilde{x} \in \varphi^{-1}(N_2).$$

Since  $\tilde{x} \in \varphi^{-1}(N_2)$ , it follows that  $\varphi(\tilde{x}) \in N_2$ . But note that  $N_2 \trianglelefteq G_2$  and so for all  $g' \in G_2$ , we have

$$g'N_2(g')^{-1} = N_2. \quad (\dagger)$$

Since  $\varphi$  is a homomorphism, we see that

$$\varphi(x) = \varphi(g\tilde{x}g^{-1}) = \varphi(g)\varphi(\tilde{x})(\varphi(g))^{-1}$$

Since  $\varphi(g) \in N_2$ , we have  $\varphi(x) \in N_2$  from  $(\dagger)$ . Thus,  $x \in \varphi^{-1}(N_2)$  and so we conclude that

$$g\varphi^{-1}(N_2)g^{-1} \subseteq \varphi^{-1}(N_2)$$

which is our desired result. We see that  $\ker(\varphi) \trianglelefteq G_1$  because

$$\begin{aligned} \ker(\varphi) &= \{g \in G_1 : \varphi(g) = 1_G\varphi^{-1}(N_2)\} \\ &= \{g \in G_1 : gN_2 = \varphi^{-1}(N_2)\} \\ &= \{g \in G_1 : g \in \varphi^{-1}(N_2)\} \\ &= \varphi^{-1}(N_2) \\ &\trianglelefteq G_1. \end{aligned}$$

■

(b) Show that if  $N_1 \trianglelefteq G_1$ , then  $\varphi(N_1) \trianglelefteq \varphi(G_1)$ . Deduce that any surjective homomorphism will map normal subgroups in the domain to normal subgroups in the codomain.

**Proof.** Suppose that  $N_1 \trianglelefteq G_1$ . To show that  $\varphi(N_1) \trianglelefteq \varphi(G_1)$ , we need to show that for all  $g' \in \varphi(G_1)$ ,

$$g'\varphi(N_1)(g')^{-1} \subseteq \varphi(N_1). \quad (*)$$

Let  $g' \in \varphi(G_1)$ . To show (\*), let  $y \in g'\varphi(N_1)(g')^{-1}$ . Then for some  $\tilde{y} \in \varphi(N_1)$ , we have

$$y = g'\tilde{y}(g')^{-1}.$$

Since  $\tilde{y} \in \varphi(N_1)$ , then for some  $\tilde{x} \in N_1$ , we have  $\tilde{y} = \varphi(\tilde{x})$ . Furthermore,  $g' \in \varphi(G_1)$  implies that there exists some  $\tilde{g} \in G_1$  such that  $g' = \varphi(\tilde{g})$ . Since  $\varphi$  is a homomorphism, it follows that

$$\begin{aligned} y &= g'\tilde{y}(g')^{-1} = \varphi(\tilde{g})\varphi(\tilde{x})\varphi(\tilde{g})^{-1} \\ &= \varphi(\tilde{g})\varphi(\tilde{x})\varphi(\tilde{g}^{-1}) \\ &= \varphi(\tilde{g}\tilde{x}\tilde{g}^{-1}). \end{aligned}$$

Notice that  $\tilde{g}\tilde{x}\tilde{g}^{-1} \in \tilde{g}N_1\tilde{g}^{-1}$ . But since  $N_1 \trianglelefteq G_1$ , we have for all  $g \in G$ ,  $gN_1g^{-1} = N_1$ . In particular,  $\tilde{g}N_1\tilde{g}^{-1} = N_1$ . Hence,  $\hat{x} = \tilde{g}\tilde{x}\tilde{g}^{-1} \in N_1$ . Hence, we can see that  $y \in \varphi(N_1)$  and so we conclude that

$$g'\varphi(N_1)(g')^{-1} \subseteq \varphi(N_1).$$

Hence, we conclude that  $\varphi(N_1) \trianglelefteq \varphi(G_1)$ .

Now, let  $\varphi : G_1 \rightarrow G_2$  be a surjective homomorphism, we see that  $\varphi(G_1) = G_2$ . Thus, the above can be written as

$$\varphi(N_1) \trianglelefteq G_2.$$

Now, if we consider  $\varphi : N_1 \rightarrow N_2$ , then  $\varphi(N_1) = N_2$  and so the above can be written further as

$$N_2 \trianglelefteq G_2.$$

Hence, any surjective homomorphism will map normal subgroups in the domain to normal subgroups in the codomain. ■

**Problem 2.** Let  $G$  be a group with  $N \trianglelefteq G$ .

- (a) Suppose  $x \in G$  with  $|x| = n < \infty$ . Show that the order of  $xN$  in  $G/N$  divides  $n$ .

**Proof.** Since  $|x| = n$ , we have  $x^n = 1_G$ . To show that  $|xN| \mid n$ , we need to show that  $(xN)^n = N$ . Since  $N \trianglelefteq G$ , we know that coset multiplication is well-defined. Hence, we see that (where the statement  $x^nN = (xN)^n$  can be proven inductively)

$$(xN)^n = x^nN = 1_GN.$$

Since the identity in  $G/N$  is  $N = 1_GN$ , it follows that  $|xN| \mid n$  and we are done. ■

- (b) Further show that the order  $xN$  in  $G/N$  is the smallest positive integer  $k$  such that  $x^k \in N$ .

**Proof.** Suppose  $N \neq \{1_G\}$ . Then there exists some  $a \neq 0$  such that  $x^a \in N$ . If  $a < 0$ , then since  $N$  is a group  $x^{-a} = (x^a)^{-1} \in N$ . This tells us that the set  $\mathcal{P} = \{b \in \mathbb{Z}^+ : x^b \in N\}$  is nonempty. Since  $\mathcal{P}$  is a subset of  $\mathbb{Z}$ , the well-ordering property implies that there exists a minimal element  $d$  such that  $x^d \in N$ . In  $G/N$ , this corresponds to

$$x^dN = 1_GN.$$

But since the order of  $xN$  in  $G/N$  is unique, this tells us that  $k = d$ . Hence,  $k$  is the smallest positive integer such that  $x^k \in N$ . ■

- (c) Show that if  $G = \langle x \rangle$ , then  $G/N$  is cyclic.

**Proof.** Since  $|x| = n$ , we know that  $|\langle x \rangle| = n$ . Also,  $|G| = |\langle x \rangle| = n$  since  $G = \langle x \rangle$ . Furthermore, notice that  $\pi : G \rightarrow G/N$  is a surjective homomorphism. Our goal is to use the quotient map to

show that  $\langle xN \rangle = G/N$ . Note that from part (a) and (b),  $|xN| \mid |x| = n$  and  $|xN| = k$  is the smallest positive integer such that  $x^k \in N$ . Hence,

$$\begin{aligned}\pi(G) &= \pi(\langle x \rangle) = \{\pi(1_G), \pi(x), \pi(x^2), \dots, \pi(x^{k-1})\} \\ &= \{1_G N, xN, x^2 N, \dots, x^{k-1} N\} \\ &= \{1_G N, xN, (xN)^2, \dots, (xN)^{k-1}\} \\ &= \langle xN \rangle.\end{aligned}$$

Thus,  $\pi(G) = \langle xN \rangle$ . But notice that  $\pi$  is surjective, we have  $\pi(G) = G/N$ . Therefore,  $G/N = \langle xN \rangle$  and so we conclude that  $G/N = \langle xN \rangle$  is cyclic. ■

**Problem 3.** Let  $G$  with center  $Z(G)$ . Recall that  $Z(G) \trianglelefteq G$ .

(a) Prove that if  $G/Z(G)$  is cyclic, then  $G$  is abelian.

**Proof.** Our goal is to show that  $G$  is abelian. Suppose  $G/Z(G)$  is cyclic. We need to show that

$$xy = yx \quad \forall x, y \in G.$$

To this end, let  $x, y \in G$  be arbitrary. Since  $G/Z(G)$  is cyclic, we must have that  $G/Z(G)$  is abelian. Since  $xy \in G$  and  $yx \in G$ , we have that (because  $Z(G) \trianglelefteq G$  and so the coset multiplication is well-defined)

$$(xy)Z(G) = (xZ(G))(yZ(G)) = (yZ(G))(xZ(G)) = (yx)Z(G).$$

This tells us that  $(xy)^{-1}(yx) \in Z(G)$  and so we have

$$(xy)^{-1}(yx)g = g(xy)^{-1}(yx) \quad \forall g \in G.$$

In particular, if  $g = x^{-1}$ , we have

$$\begin{aligned}(xy)^{-1}(yx)x^{-1} &= x^{-1}(xy)^{-1}(yx) \\ \implies y^{-1}x^{-1}y &= x^{-1}y^{-1}x^{-1}yx \\ \implies x(y^{-1}x^{-1}y) &= (y^{-1}x^{-1}y)x.\end{aligned}$$

Since  $x \in G$  was arbitrary, we see that  $y^{-1}x^{-1}y \in Z(G)$ . That is,

$$(y^{-1}x^{-1}y)g = g(y^{-1}x^{-1}y) \quad \forall g \in G.$$

In particular, if  $g = y$ , then

$$\begin{aligned}y(y^{-1}x^{-1}y) &= (y^{-1}x^{-1}y)y \\ \implies x^{-1}y &= (y^{-1}x^{-1}y)y \\ \implies x^{-1} &= y^{-1}x^{-1}y \\ \implies xy &= yx.\end{aligned}$$

Hence,  $G$  must be abelian. ■

(b) Show that if  $|G| = pq$  where  $p$  and  $q$  are primes (not necessarily distinct) then either  $G$  is abelian or  $Z(G) = \{1_G\}$ .

**Proof.** Suppose  $|G| = pq$  where  $p$  and  $q$  are primes. Suppose  $Z(G) \neq \{1_G\}$ . Our goal is to show that  $G$  is abelian. Since  $G$  is a finite group and  $Z(G) \leq G$ , it follows from Lagrange's Theorem that

$$|Z(G)| \mid |G|.$$

This tells us that  $|Z(G)| \in \{1, p, q, pq\}$ . If  $|Z(G)| = 1$ , then we would get a contradiction because we assumed that  $Z(G) \neq 1$ . If  $|Z(G)| = p$ , then

$$|G/N| = \frac{|G|}{|Z(G)|} = \frac{pq}{p} = q.$$

Since  $q$  is prime,  $G/N$  must be a cyclic group of order prime  $q$ . From part (a),  $G$  must be abelian. Similarly, if  $|Z(G)| = q$ , then we would get  $|G/N| = p$ , and so  $G/N$  is a cyclic group of order prime  $p$  and hence  $G$  is an abelian group. If  $|Z(G)| = pq = |G|$  and  $Z(G) \leq G$ , then  $Z(G) = G$  and we also get that  $G$  is abelian.

If  $p$  and  $q$  are not distinct, then  $|Z(G)| \in \{1, p, p^2\}$ . A similar argument will show that  $G$  will be an abelian group given that  $|G/N|$  is prime. For the case that  $|Z(G)| = p^2$ , then  $Z(G) = G$  and so  $G$  is abelian. ■

**Problem 4** (Section 3.2 Problem 5). Let  $H$  be a subgroup of  $G$  and fix some element  $g \in G$ .

- (a) Prove that  $gHg^{-1}$  is a subgroup of  $G$  of the same order as  $H$ .

**Proof.** Let  $g \in G$ . We will prove that  $gHg^{-1} \leq G$  and that  $|gHg^{-1}| = |H|$ . Note that  $gHg^{-1}$  is nonempty because  $1_G \in gHg^{-1}$ . Indeed, since  $H$  is a subgroup of  $G$ , we know that  $1_G \in H$  and so

$$g1_Gg^{-1} = gg^{-1} = 1_G.$$

Hence,  $1_G \in gHg^{-1}$ .

Now, let  $x, y \in gHg^{-1}$ . Then we have

$$x = gh_1g^{-1} \text{ and } y = gh_2g^{-1} \text{ for some } h_1, h_2 \in H.$$

Then we have

$$xy = (gh_1g^{-1})(gh_2g^{-1}) = g(h_1h_2)g^{-1}$$

We know that  $h_1h_2 \in H$  because  $H$  is a subgroup of  $G$  and it is closed under multiplication. Hence,  $xy \in gHg^{-1}$ . Since  $H$  is also closed under inverses, we have that  $h_1^{-1} \in H$ . Therefore, we have

$$x^{-1} = (g(h_1g^{-1}))^{-1} = (h_1g^{-1})^{-1}g^{-1} = gh_1^{-1}g^{-1}.$$

Thus, we can see that  $x^{-1} \in gHg^{-1}$  and so  $gHg^{-1}$  is also closed under inverses. This tells us that  $gHg^{-1} \leq G$ .

To show that  $gHg^{-1}$  and  $H$  have the same order, we will show that the mapping  $\varphi : H \rightarrow gHg^{-1}$  defined by  $\varphi(h) = ghg^{-1}$  for all  $h \in H$  is bijective. Indeed, we can see that  $\varphi$  is injective because for  $h_1, h_2 \in H$  such that  $\varphi(h_1) = \varphi(h_2)$ , we have

$$\begin{aligned} gh_1g^{-1} &= gh_2g^{-1} \\ gh_1 &= gh_2 \\ h_1 &= h_2 \end{aligned}$$

by multiplying by  $g$  on the right and  $g^{-1}$  on the right and left, respectively. Hence, we see that  $\varphi$  is an injective map. Also, it is immediate that  $\varphi$  is surjective. Indeed, for every  $y \in gHg^{-1}$ , we see (by definition of  $gHg^{-1}$ ) that there exists some  $\hat{h} \in H$  such that  $y = g\hat{h}g^{-1} = \varphi(\hat{h})$ . Hence,  $\varphi$  is a bijective mapping from  $H$  to  $gHg^{-1}$ . Thus, we conclude that  $|H| = |gHg^{-1}|$ . ■

(b) Deduce that if  $n \in \mathbb{Z}^+$  and  $H$  is the unique subgroup of order  $n$  then  $H \trianglelefteq G$ .

**Proof.** Because  $|gHg^{-1}| = |H|$  for all  $g \in G$  from the previous part and that  $H$  is a unique subgroup of order  $n$ , it must follow that  $gHg^{-1} = H$  and so  $H \trianglelefteq G$ . Otherwise, if there exists some other subgroup, say  $H'$ , of the same order as  $H$ , then  $|H'| = |H| = |gHg^{-1}|$  but necessarily  $H \neq H' = gHg^{-1}$ . ■

**Problem 5** (Section 3.2 Problem 8). Prove that if  $H$  and  $K$  are finite subgroups of  $G$  whose orders are relatively prime then  $H \cap K = 1$ .

**Proof.** Suppose  $H$  and  $K$  are finite subgroups of  $G$  whose orders are relatively prime. Our goal is to show that  $H \cap K = \{1_G\}$ . To this end, it suffices to show that  $|H \cap K| = 1$ . Note that if  $H$  and  $K$  are trivial subgroups, then the result follows immediately and so suppose otherwise. Since  $|H|$  and  $|K|$  are relatively prime, we know that  $(|H|, |K|) = 1$ . Hence,  $1 \mid |H|$  and  $1 \mid |K|$  and so we have  $1 \mid |H||K|$ . Because  $H$  and  $K$  are subgroups of finite order, we have

$$|HK| = \frac{|H||K|}{|H \cap K|} \iff |HK||H \cap K| = |H||K|.$$

Because  $1 \mid |H||K|$ , either  $|HK| = 1$  or  $|H \cap K| = 1$ . If  $|HK| = 1$ , then  $|H|$  and  $|K|$  must both be equal to 1. However, we assumed that  $H$  and  $K$  are non-trivial subgroups of  $G$  and so they must have more elements other than the identity. Thus, they cannot be both equal to 1 which is a contradiction. Hence, it must be that  $|H \cap K| = 1$  and so we conclude that  $H \cap K = \{1_G\}$ . ■